

COURSE TITLE : SEVER AND NETWORK ADMINISTRATION
COURSE CODE : 5261
COURSE CATEGORY : A
PERIODS/WEEK : 4
PERIODS/SEMESTER : 52
CREDITS : 4

TIME SCHEDULE

MODULE	TOPICS	PERIODS
1	Unix Basics	13
2	UNIX Server Configurations	13
3	Network Management	13
4	Internetworking	13

COURSE GENERAL OUTCOMES:

Sl.	G.O	On completion of this course the student will be able to :
1	1	Comprehend the basic commands, filesystem and utilities of unix
2	1	Deploy different servers in Unix based systems
3	1	Understand networking basics and Network management commands
4	1	Demonstrate internetworking concepts

SPECIFIC OUTCOMES:

MODULE – I: Unix Basics

1.1 Comprehend the basic commands, filesystem and utilities of unix

- 1.1.1 To explain about command line and shell.
- 1.1.2 To apply different UNIX commands like man,date,who,pwd,cat.
- 1.1.3 To use different editors such as gedit, vi editor etc.
- 1.1.4 To describe unix file system and apply file handling utilities
- 1.1.5 To use basic file & directory management commands
- 1.1.6 To apply the concept of file permissions for security.
- 1.1.7 To use su and sudo command

- 1.1.8 To demonstrate user management in Unix

MODULE – II: UNIX Server Configurations

2.1 Deploy different servers in Unix based systems

- 2.1.1 To use yum utility for package installation.
- 2.1.2 To use rpm for new package installation.
- 2.1.3 To employ apache HTTP server in Unix based machines
- 2.1.4 To deploysquid, dhcp server , samba and ftp in Unix based machines

MODULE – III: Network Management

3.1 Understand networking basics and Network management commands

- 3.1.1 To summarize TCP/IP basics
- 3.1.2 To analyse IP addressing and subnetting.
- 3.1.3 To employ Client server mechanism.
- 3.1.4 To explain Network management commands such as hostname, ifconfig, finger, talk, Ping, Traceroute, Arp, Route, Netstat, telnet, ftp, dnsetc
- 3.1.5 To apply and use Network management commands such as hostname, ifconfig, finger, talk, Ping, Traceroute, Arp, Route, Netstat, telnet, ftp, dnsetc

MODULE – IV: Internetworking

4.1 Demonstrate internetworking concepts

- 4.1.1 To identify and describe nic, hub, bridge, switch, router and different types of cables
- 4.1.2 To recognize color coding for various types of crimping of network cable
- 4.1.3 To demonstrate and apply crimping
- 4.1.4 To summarise LAN and WAN
- 4.1.5 To apply basic router configuration commands
- 4.1.6 To practice and apply static routing
- 4.1.7 To demonstrate and apply dynamic routing
- 4.1.8 To construct simple WAN scenario

CONTENT DETAILS

MODULE – I: Unix Basics

Accessing the Command Line/Shell, Simple commands-man command,date,who,pwd,cat, gedit,vi editor, Introduction to unix file system, file handling utilities,Basic File & Directory

management commands – cd,ls,cp,mv,mkdir,rm,head,tail, security by file permissions, su and sudo commands, User management

MODULE – II: UNIX Server Configurations

Yum, rpm utilities, configurations of apache HTTP server, squid, dhcpserver , samba and ftp

MODULE – III: Network Management

TCP/IP Basics, IP addressing and subnetting, Client server mechanism, Network management commands – hostname,ifconfig,finger,talk,Ping, Traceroute, Arp, Route, Netstat, telnet, ftp, dns

MODULE – IV: Internetworking

Network hardware:- nic, hub, bridge, switch, router and different types of cables, color coding, crimping RJ 45, Overview of LAN and WAN, Router configuration basics:- basic commands, static routing, dynamic routing using RIP, Implementation of Simple WAN scenario.

TEXT BOOK(S)

1. Behrouz A. Forouzan, Richard F. Gilberg, "*Unix and shell Programming.*", Cengage Learning.
2. SumitabhaDas , "*Unix the ultimate guide*", TMH. 2nd Edition.
3. Behurouz&Forozan, "*Introduction to Data Communications & Networking*" –Mc Graw Hill

REFERENCE

1. Kernighan and Pike, "*Unix programming environment*", PHI. / Pearson Education
2. Graham Glass, King Ables, "*Unix for programmers and users*", 3rd edition, PearsonEducation
3. Maurice J. Bach, "*The Design of the Unix Operating System*", First Edition,Pearson Education, 1999
4. A.S. Tanenbaum, "*Computer Networks*"– PHI
5. William Stallings, "*Data and Computer Communication*" - Pearson EducationAsia
6. https://access.redhat.com/documentation/enUS/Red_Hat_Enterprise_Linux/7/html/System_Admistrators_Guide